Week 3: Development of Data Visualizations in Looker Studio

Data Visualization Associate Internship

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EXCELERATE – POWERED BY SAINT LOUIS UNIVERSITY (SLU) INTERNSHIP PROGRAM

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1 INTRODUCTION

Week 3 of the internship focused on preparing the foundation for building a meaningful dashboard. After completing the data cleaning and ETL process in Week 2, this phase aimed to ensure the dataset was ready for visualization. The focus was on understanding the cleaned Master Table, creating a precise mapping between data columns and dashboard metrics, and designing visual wireframes that outline how the final dashboard would look. These steps are essential to support insightful data storytelling and effective decision-making.

1.1 Motivation

Raw data, even if cleaned, is only valuable when converted into actionable insights. Dashboards make this possible by visually summarizing complex datasets into digestible information. The motivation behind this week's work was to bridge the gap between data and insights by designing a thoughtful structure for dashboard development — making it easier for stakeholders to interpret and act on data.

1.2 Report Overview

This report documents the key activities and outcomes of Week 3. It covers the understanding and analysis of the cleaned Master Table, the creation of a detailed Mapping Table, and the development of annotated wireframes for two dashboards — one for the Master Table and one for the marketing dataset. Each task aligns with preparing a comprehensive, insightful dashboard in Week 4.

1.3 Problem Statement

While a cleaned dataset provides a reliable base, it still needs to be connected meaningfully to dashboard metrics and visuals. Without a proper mapping and visual layout plan, dashboard creation can be disorganized, inconsistent, or misleading. There was a need to:

- Translate cleaned data into dashboard-ready formats.
- Align data fields to KPIs, dimensions, and filters.
- Design wireframes to define the dashboard structure and layout.
- Handle additional datasets like marketing1.csv which were not part of the Master Table.

1.4 Objective

- Understand the structure and context of the cleaned Master Table.
- Create a detailed Mapping Table to link dashboard fields with source columns.
- Define and organize key KPIs, metrics, and filters for dashboard presentation.
- Design clear and annotated wireframes for both the Master and Marketing datasets.
- Set up a visual and logical blueprint for building the final dashboard in Week 4.

2 MAPPING TABLE CREATION PROCEDURE

Creating a **Mapping Table** is a critical part of dashboard preparation. It aligns the fields from cleaned Master Table with how they will be used in the final visualization. This ensures that the dashboard reflects accurate KPIs, dimensions, filters, and business rules — and it sets a solid foundation for Week 4.

2.1 Understanding the Master Table

The Master Table, cleaned and finalized in Week 2, integrates key information from five datasets:

- learneropp1 (enrollment data)
- learner1 (user demographic data)
- cohort1 (program/cohort information)
- opportunity1 (course/opportunity details)
- cognito2 (login/identity data)

Each column in this unified table holds meaningful information about the learners, their enrollment, program cohorts, and opportunity details. Before mapping, a thorough review was performed to understand what each column represents. The table captures demographic, academic, and engagement-related information.

2.2 Identify Important Columns

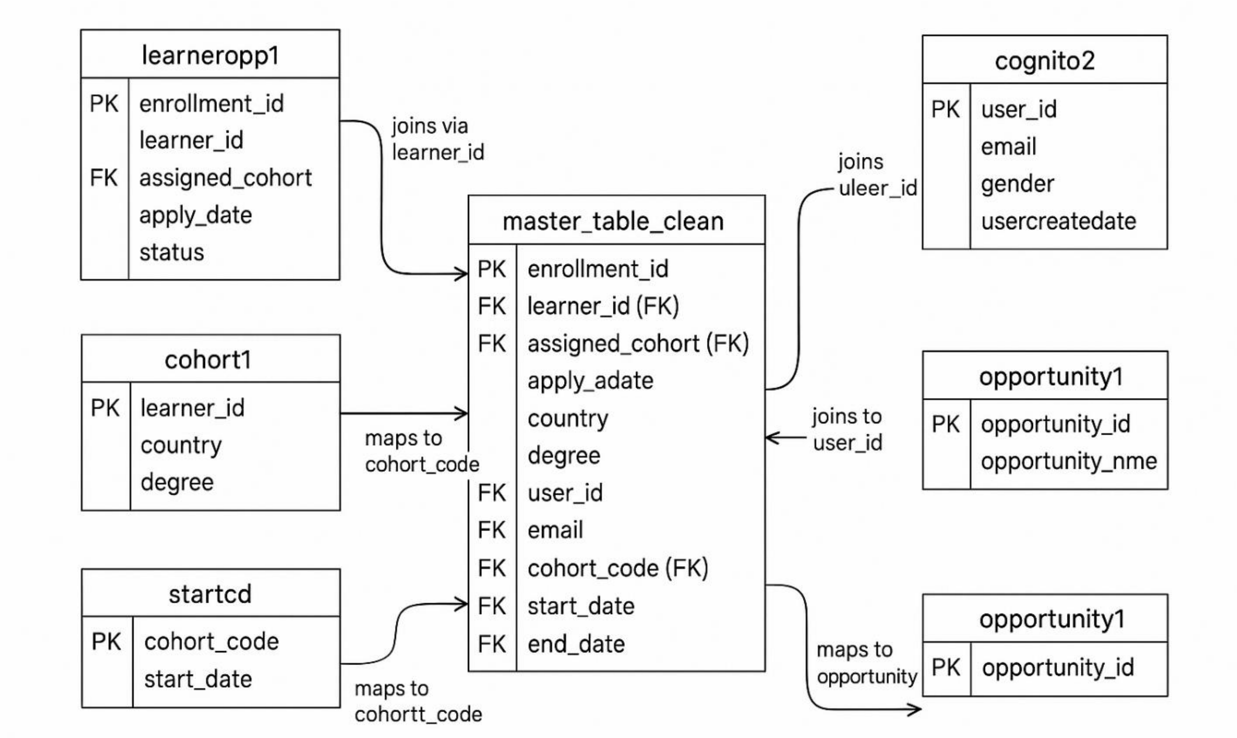
We classified columns into three categories for dashboard usage:

- **KPIs (Key Performance Indicators)**:
  Metrics that summarize performance — e.g., total learners, dropout, average performance score, active opportunities.
- **Supporting Metrics**:
  Data that supports deeper analysis — e.g., cohort duration, number of programs, region-based stats.
- **Dimensions (Categories)**:
  Fields used for filtering, grouping, or segmentation — e.g., country, gender, degree, cohort code, status.

2.3 Understand Column Relationships

Relationships were established as follows:

- learner_id links personal/demographic data from learner1, cognito2, and enrollment details.
- assigned_cohort maps to cohort_code in cohort1 for program tracking.
- opportunity_id links to program/course names and categories in opportunity1.



These relationships ensure that we can combine visual and demographic context, track learner progression, and apply filters for meaningful insights.

2.4 Designing the Mapping Table in Excel

Below is the **actual structure** of the Mapping Table created in Excel. It contains each dashboard field, its source column, datatype, transformation (if any), and how it will be used in the dashboard:

Dashboard Field	Source Column (Master Table)	Data Type	Transformation Applied	Business Logic / Purpose
Enrollment ID	enrollment_id	Text	None	Unique record identifier
Country	country	Text	Standardized capitalization	Used as a filter
Cohort Code	cohort_code	Text	None	Drill-down and grouping
Gender Distribution	gender	Text	Normalized values (Male/Female)	KPI box
Birth Year	birthdate	Date	Extract year using formula	Used to calculate age distribution
Program Name	opportunity_name	Text	Cleaned inconsistent naming	Main category field

Cohort Duration	start_date, end_date	Date	Calculated duration using DATEDIFF	Used in trend and duration charts
Completion Status	status	Numeric	Flag 1070 as Completed	KPI and filter
Email Count	email	Text	NULL values removed	For user engagement metrics

2.5 Final Mapping Table and Dashboard Readiness

After completing the mapping:

- All transformations and logic were documented inside the Excel file.
- The table includes **clear traceability** from raw to cleaned to dashboard-ready columns.
- This file supports dashboard developers in understanding which fields to use and how to represent them.

The final Mapping Table was saved as an Excel file and submitted for review.

2.6 🗀 Mapping Table Drive Link:

You can view the complete Mapping Table created for this project using the following link:

🔗 **Download the Mapping Table (Click Here)**

3 Dashboard Wireframe Planning and Design

This section outlines the planning and design of the dashboard wireframe based on the cleaned Master Table from Week 2. The wireframe serves as a **blueprint** for building the final dashboard in Week 4, ensuring clarity, usability, and alignment with key metrics and business logic.

3.1 Why Wireframing is Essential

Before constructing the final dashboard, creating a wireframe is vital because it:

- Visually **structures your thoughts** and approach.
- Ensures **critical KPIs and insights** are well-positioned.
- Helps build a **user-friendly and intuitive interface**.
- **Prevents redesign** later by setting a solid foundation.
- Allows for **early feedback** and iteration from stakeholders.

The wireframe acts as a **visual prototype**, showing what elements (KPI cards, graphs, filters) will appear where and why.

3.2 Key Dashboard Sections

Our dashboard is structured into clearly defined areas:

KPI Boxes (Top Row)

- Placed at the top of the dashboard for **quick performance tracking**.
- Usually shows 3–4 key metrics such as:
  - Total Enrollments
  - Dropout Rate
  - Average Completion Rate
  - Active vs Inactive Learners

Graph Placements (Middle Section)

- Main visualizations go here.
- Example chart types:
  - **Bar Charts**: Dropout Rate by Course, Enrollment by Country
  - **Line Charts**: Monthly Application Trends
  - **Pie Charts**: Learner Gender Distribution or Degree Breakdown
  - **Tables**: List of top-performing cohorts, learners, or institutions

Filters and Drill-Down Controls

Allow us to filter data by:

- Department
- Course Type
- Gender
- Country
- Time Range

Drill-down: e.g., clicking on a course shows learner-level detail.

3.3 Wireframe Design Explanation(Master Table)

For guidance of our Week 4 dashboard, a detailed wireframe was created, reflecting the core KPIs, charts, filters, and interactive elements.

At the top of the layout, we placed **four KPI cards** to provide a high-level summary:

- **Total Learners** offers a count of unique users on the platform.
- **Dropout Rate** serves as a red flag metric to monitor retention.
- **Average Performance** highlights learner quality and academic strength.
- **Active Opportunities** gives insight into the scale of available resources for learners.

The **middle section** consists of three strategic visuals:

- A **bar chart** showing **Dropout Count by Cohort** helps identify underperforming batches.
- A **line chart** tracking the **Enrollment Trend Over Time** is useful to observe seasonal spikes or declines in interest.
- A **pie chart** illustrating **Learner Distribution by Country** informs marketing or regional strategy decisions.

The **filter panel**, placed along the side or top-right, allows users to narrow their view using:

- **Cohort Code**, **Gender**, **Status**, and **Country**.

At the bottom of the wireframe, we included a **detailed data table** that allows drill-down into individual learners' profiles, performance, and opportunity engagement. This table supports granular insights not visible in summarized KPIs.

Annotations were added to the wireframe image to justify the placement, chart types, and design logic, ensuring clarity and transparency in our layout.

3.4 Marketing Dataset

In order to assess and visualize the effectiveness of our marketing efforts, a dedicated wireframe was designed for the **marketing1 dataset**. This dashboard focuses on spend efficiency, audience engagement, and campaign performance.

Wireframe Design Explanation – Marketing Dashboard

The **KPI section at the top** includes:

- **Total Campaigns** to summarize how many initiatives are running.
- **Total Amount Spent (AED)** to show the investment made.
- **Total Outbound Clicks** indicating audience interaction.
- **Average Cost per Result** providing insight into the efficiency of each marketing effort.

In the **chart section**, we included:

- **Total Results by Campaign (bar chart)** to compare outcomes of different campaigns.
- **Cost per Result by Campaign (bar chart)** to evaluate which campaigns are cost-efficient.
- **Reach vs Outbound Clicks (scatter plot)** to explore how audience exposure relates to user engagement.

The **filter panel** enables slicing data by:

- **Campaign Name**, **Delivery Status**, and **Reporting Period**, so users can focus on specific timeframes or performance stages.

At the bottom, we added a **campaign-level table** summarizing key metrics like cost, reach, results, and CPC for granular performance reviews.

Each section of the wireframe is annotated to explain **what** it shows and **why** it is useful for marketing optimization. This structured approach ensures clarity, usability, and strategic insights for stakeholders.

Tool Used

We used **Figma** for creating wireframe of both dashboard, **main dashboard(Master Table)** & **Marketing Dataset.**

3.5 🗀 Wireframe Dashboard Drive Link:

🔗 Download Wireframe PDF (Click here)

Summary of Week-3:

- Reviewed the cleaned Master Table thoroughly to understand each column's meaning and relevance for dashboard use.
- Identified and categorized key data fields as KPIs, metrics, and dimensions (filters).
- Created a detailed Mapping Table in Excel including dashboard field names, corresponding source columns, data types, applied transformations, and business logic.
- Ensured all transformations such as date formatting, text normalization, and null handling were documented for visualization readiness.
- Designed the main dashboard wireframe layout using a clean and structured format with sections for KPIs, graphs, filters, and data tables.
- Included annotations on the wireframe explaining each element's role — why KPIs were chosen, chart types used, and the purpose of filters.
- Created a separate wireframe for marketing data covering campaign-based metrics, performance visuals, and relevant filters.
- Final outputs include both Mapping Table (Excel) and annotated wireframes (PNG), ready for use in Week 4 dashboard development.

4 CONCLUSION

In Week 3, we laid a strong foundation for effective dashboard development by strategically mapping and visualizing our data. Through the Mapping Table, we aligned each key dashboard field with its source column from the cleaned Master Table, clearly defining data types, transformations, and business logic. This ensures consistency and accuracy in our visuals.

The wireframe design process allowed us to thoughtfully plan the layout and structure of our dashboard before implementation. By identifying key KPIs, choosing the right chart types, and integrating filters, we focused on usability and clarity for end-users. Annotating each component further clarified its purpose and analytical value.

Additionally, since the marketing dataset was not part of the Master Table, a separate wireframe was designed for it. This focused on campaign performance KPIs like *Total Spend*, *Total Results*, *Cost per Result*, and *Click-Through Rates*. Charts like *Cost per Campaign*, *Reach vs Clicks*, and *Results by Campaign* were mapped out with clear annotations for marketing insights.

Overall, Week 3 provided a strategic bridge between data cleaning and dashboard building**,** equipping us with a clear, structured plan to create an insightful, interactive, and user-friendly dashboard in Week 4. Both the Mapping Table and the Wireframes will serve as the backbone of our visualization journey moving forward for week 4.